ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** the container and the flame also differ along with the volume	Volume of water is a variable.
2	**(d)** a grass	Grasses show these features.
3	**(a)** starch and fat	Both tests are positive.
4	**(d)** P and R are magnets, Q is magnetic material	Repulsion is seen only between magnets.
5	**(b)** 160 cm	320 - 160.
6	**(a)** P	The pin falls first.
7	**(b)** 50 °C	Linear scale with equal spacing gives 50 °C.
8	**(b)** heat is used to change the state	Latent heat idea.
9	**(a)** magnet, add water and filter, evaporate filtrate	Each step removes one component.
10	**(d)** it does not grow or reproduce on its own	Many characteristics are absent.
11	**(d)** most oxygen was used	Burning consumed oxygen.
12	**(d)** 8 light-minutes	Approx.
13	**(a)** both depend on hot weather	A third factor links them.
14	**(b)** helps judge distance for hunting	Forward eyes give depth perception.
15	**(d)** they contain much sugar and few nutrients	Empty calories.
16	**(a)** iron is magnetic and copper is not	Only iron is attracted.
17	**(a)** Both A and R are true and R is the correct explanation of A.	A hypothesis is a testable explanation, and testing it by an experiment is exactly how we check it.
18	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R is a general pattern and does not explain why wheat has fibrous roots.
19	**(a)** Both A and R are true and R is the correct explanation of A.	Growth and repair of tissues is why proteins are called body-building nutrients.
20	**(a)** Both A and R are true and R is the correct explanation of A.	The Earth's magnetic field acts on the suspended magnet and aligns it, so R explains A.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Two factors differ: light and water (1); We cannot tell which factor caused the difference (1).	1+1
22	Green stem carries out photosynthesis (1); Spines reduce water loss (1).	1+1
23	It contains starch (1); It also contains fat (1).	1+1
24	The steel table disturbs the needle (1); Move it to a wooden or non-magnetic surface (1).	1+1
25	3.6 km = 360000 cm (1); 360000 / 60 = 6000 steps (1).	1+1
26	Empty bottle has low average density (1); Sand makes it heavier (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Use same size, same wetness (1); Same place, time and temperature (1); Measure drying time for each (1).	1+1+1
28	Tree (1); Tap root goes deep (1); Net veins in leaves (1).	1+1+1
29	Lacks protein and vitamins (1); High in refined carbohydrates and fat (1); Add milk, dal, vegetables and fruit (1).	1+1+1
30	Roll bar in filings (1); Maximum cling near the ends (1); Little in the middle (1).	1+1+1
31	1.1 km = 1100 m (1); 1100 / 2.2 (1); 500 turns (1).	1+1+1
	OR: It is a metal (1); Shiny (lustrous) and sinks (1); Conducts heat along its length (1).	1+1+1
32	Atmospheric pressure is lower on hills (1); Water boils at a lower temperature (1); Therefore reading differs (1).	1+1+1

	OR: 4 x 30 = 120 L (1); 120/2000 (1); 6 per cent (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) A hypothesis (1). (ii) The amount of sunlight (1). (iii) Recording the observations (or results) (1). (iv) Only one thing should change so that any difference can be blamed on sunlight (1); the plant in sunlight grew better, so plants need sunlight (1).	1+1+1+2
34	(i) Mango (1). (ii) Pumpkin (1). (iii) Herb: small with soft, green stem; shrub: bushy with a hard stem branching near the base (1). (iv) A creeper spreads on the ground with a weak stem (1); a climber climbs up a support with tendrils or other parts; example: grapevine (1).	1+1+1+2
35	A: vitamin C, citrus fruits (1); B: vitamin A, carrot (1); C: iodine, iodised salt (1); D: vitamin D, sunlight and milk (1); E: iron, spinach (1).	1+1+1+1+1
36	Same type of clips (1); Same method to approach (1); Count clips lifted by each pole (1); Repeat three times (1); Compare averages (1).	1+1+1+1+1
	OR: Tree trunk: measuring tape or thread around (1); Then measure thread (1); Coin: stack 20 coins and measure (1); Divide by 20 (1); Pencil: ruler with zero at one end (1).	1+1+1+1+1
37	Same heat source and time (1); Measure temperature of the free end (1); Lowest rise is best (1); Use tongs and avoid burns (1); Rank and justify (1).	1+1+1+1+1
	OR: Same volume, same starting temperature (1); Same containers (1); Record temperature every 2 min (1); Plot a graph or table (1); Compare cooling (1).	1+1+1+1+1
38	Identical wet cloths (1); Place in three locations (1); Weigh at intervals (1); Greatest loss shows the fastest drying (1); Conclusion with reasons (1).	1+1+1+1+1
	OR: Keep both still (1); Mud settles, salt water stays clear (1); Filter both (1); Mud on paper, salt in filtrate (1); Evaporate filtrate for salt (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 30 g	Same as initial.
2	**(a)** pebble	It has one cell.
3	**(d)** adding compost	Compost adds humus.
4	**(b)** 60 kg	Mass does not change.
5	**(d)** compare the methods and repeat carefully	Differences in method may explain.
6	**(a)** many plants will fail to pollinate	Pollination would fall.
7	**(b)** fat	Protein test.
8	**(a)** always in the same direction with the same pole of a magnet	Opposite pole forms at the lifting end.
9	**(d)** 10.5 cm	12.5 - 2.
10	**(d)** its shape displaces much water	Hollow shape lowers average density.
11	**(a)** aluminium and plastic	Pan conducts, handle insulates.
12	**(c)** it changed to vapour and mixed with air	Matter is conserved.
13	**(d)** it makes tiny particles clump and settle	Coagulation.
14	**(d)** a living thing can stay inactive until conditions suit it	Life processes were only slowed down.
15	**(b)** more public transport	Fewer vehicles.
16	**(b)** Earth rotates on its axis	Tilt.
17	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the conversion does not explain why metre is the SI unit.
18	**(a)** Both A and R are true and R is the correct explanation of A.	We can see clearly through glass because light passes through it; R explains A.
19	**(a)** Both A and R are true and R is the correct explanation of A.	After holding a hand in cold water, lukewarm water feels warm; this shows touch can mislead, which R explains.
20	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the fixed boiling temperature does not explain why evaporation occurs at every temperature.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Temperatures are equal (1); The larger quantity contains more heat energy (1).	1+1
22	Vapour cannot escape (1); Air in the bag becomes saturated (1).	1+1
23	Some salt stuck to the filter or vessel (1); Handling losses (1).	1+1
24	It does not grow or reproduce (1); It does not respire or respond (1).	1+1
25	Oxygen about 21 per cent is used (1); Water enters to fill that space (1).	1+1
26	228/150 (1); about 1.5 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Magnet for iron (1); Water: wooden chips float, skim (1); Filter sand and chalk; evaporate salt (1).	1+1+1
28	No chlorophyll (1); Do not make their own food (1); Absorb food from dead matter (1).	1+1+1
29	21 per cent of 50 (1); 0.21 x 50 (1); 10.5 m^3 (1).	1+1+1
30	5000/10 (1); 500 days (1); Year/day (1).	1+1+1
31	Curiosity: why do things fall (1); Observation: fall of apple and Moon's motion (1); Reasoning: same force acts on both (1).	1+1+1

	OR: Colour can vary within a group (1); Unrelated animals may have the same colour (1); Features like body covering and habitat are better (1).	1+1+1
32	Calcium for bones (1); Proteins for growth (1); Vitamins like A and D (1).	1+1+1
	OR: Permanent magnet is present (1); A coil moves in its field (1); Produces sound (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) Protein (1). (ii) Vitamins (and minerals) (1). (iii) The ingredients and nutrients in the food (and its date of expiry) (1). (iv) No single food gives all the nutrients (1); a mix gives carbohydrates, proteins, vitamins and minerals in balance, while junk food has too much fat, sugar and salt (1).	1+1+1+2
34	(i) North (and south) direction (1). (ii) North pole and south pole (1). (iii) The magnet's poles attracted or repelled the needle, which is itself a small magnet (1). (iv) Its north-pointing needle shows north; then south, east and west can be found (1), so travellers can find the way even where there are no landmarks (1).	1+1+1+2
35	Choose a scale for sizes (1); Place the Sun at the centre (1); Planets in order (1); Label features (1); Show orbits (1).	1+1+1+1+1
36	Two sets of same plants (1); One with music, one without (control) (1); Same water, soil, light (1); Measure height weekly (1); Compare and conclude (1).	1+1+1+1+1
	OR: Animals lose food and shelter (1); Plant cover declines (1); Plant more trees (1); Create national parks (1); Avoid pollution and hunting (1).	1+1+1+1+1
37	Cereal for energy (1); Dal for proteins (1); Vegetables for vitamins (1); Curd for calcium (1); Salad and pickle for fibre and flavour (1).	1+1+1+1+1
	OR: Roll magnet in filings (1); Filings gather at the ends (1); Fewer in middle (1); Count or compare quantity (1); Poles are strongest (1).	1+1+1+1+1
38	One reading may have error (1); Repeat several times (1); Take average (1); Avoid parallax (1); Use a good scale (1).	1+1+1+1+1
	OR: Take a steel foil (1); Place flat: sinks (1); Shape as a boat (1); It floats (1); Because shape holds air (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 850 m	2700 - 1850.
2	**(a)** S	The pin falls last.
3	**(d)** metal cup	Metal conducts heat away fastest.
4	**(c)** melting and boiling	States change.
5	**(d)** sand	Insoluble.
6	**(a)** the crystal grows by outside deposition, not from within	Non-living.
7	**(c)** oxygen forms about one-fifth of air	Oxygen is used up.
8	**(a)** waxing	Increasing.
9	**(d)** amount of water	Water amount varies.
10	**(a)** wide field of view to spot predators	Wide view.
11	**(c)** protein and fat	Plant foods have less protein and fat.
12	**(b)** iron only	Only iron is magnetic.
13	**(a)** it is in linear motion and its speed increases	It falls along a straight line, speeding up.
14	**(a)** C conducts heat and D does not	Test for conduction.
15	**(a)** the 20 °C object	Heat enters the colder one.
16	**(c)** 10 mL	50/5.
17	**(a)** Both A and R are true and R is the correct explanation of A.	The difference in weight is used with wind, so R explains A.
18	**(a)** Both A and R are true and R is the correct explanation of A.	Not showing life processes is why a stone is non-living.
19	**(a)** Both A and R are true and R is the correct explanation of A.	Because it is continuously available, sunlight is renewable.
20	**(a)** Both A and R are true and R is the correct explanation of A.	A star makes its own light and heat, so R explains why the Sun is a star.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	It represents the Earth in small size (1); It helps us study things too large to observe wholly (1).	1+1
22	It breathes air through lungs (1); It gives birth to young and feeds milk (1).	1+1
23	Lacks carbohydrates and fibre (1); Lacks many vitamins (1).	1+1
24	No (1); Heat weakens the magnet (1).	1+1
25	0.003 km = 3 m (1); 3 m = 300 cm (1).	1+1
26	Rubber is an electrical insulator (1); Prevents electric shock (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Use a flexible tape (1); Follow the curve (1); Verify with thread (1).	1+1+1
28	Prevent shock (1); Rubber or plastic (1); Insulate the hand (1).	1+1+1
29	Reduces loss of hot air (1); Stops steam escaping (1); Keeps heat (1).	1+1+1
30	Moist air rises over hills (1); It cools and condenses (1); Clouds drop rain (1).	1+1+1
31	Immiscible (1); Kerosene lighter (1); Floats on top (1).	1+1+1
	OR: Respiration is very slow (1); It germinates later (1); So it is alive but dormant (1).	1+1+1
32	Half the sheets (1); 100 sheets (1); 100 saved (1).	1+1+1

	OR: Gravity pulls planets (1); Pull bends their path (1); Without it they would fly off in straight lines (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) Footstep (1). (ii) Their steps are of different sizes (1). (iii) Metre (1). (iv) 9 m = 900 cm (1); a tape gives the same value for everyone because it uses a standard unit (1).	1+1+1+2
34	(i) Steel (a metal) (1). (ii) Glass (1). (iii) Wood (1). (iv) Metal conducts heat well, so it heats the food (1); wood is a poor conductor so the handle stays cool to hold (1).	1+1+1+2
35	Sea water collects in pans (1); Evaporation in stages (1); Brine thickens (1); Crystals formed (1); Washed and dried (1).	1+1+1+1+1
36	Roots to water (1); Shoots to light (1); Camouflage (1); Fleeing predators (1); Hibernation or migration (1).	1+1+1+1+1
	OR: Reduce plastic (1); Walk or cycle (1); Save water (1); Recycle (1); Plant trees (1).	1+1+1+1+1
37	Moon's phase depends on position (1); Not on place (1); Rise times differ from place to place (1); Earth rotates (1); The same phase is seen everywhere (1).	1+1+1+1+1
	OR: Sum = 90 s (1); Average = 30 s (1); One swing = 1.5 s (1); Timing many swings reduces error in starting and stopping (1); Dividing gives the time for one swing (1).	1+1+1+1+1
38	Hollow bones: lightness (1); Feathers: flight and warmth (1); Beak: food type (1); Feet: perching, swimming or hunting (1); Examples: duck, eagle, parrot (1).	1+1+1+1+1
	OR: Breakfast: milk, fruit, paratha (1); Lunch: roti, dal, vegetable, curd (1); Snack: sprouts or fruit (1); Dinner: light khichdi with vegetables (1); Avoid chips, soda and excess sweets (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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